

Electrolytes Enriched by Crown Ethers for Lithium Metal Batteries

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Lithium (Li) metal battery is considered the most promising next-generation battery due to its low potential and high theoretical capacity. However, Li dendrite growth causes serious safety problems. Herein, the 15-Crown-5 (15-C-5) is reported as an electrolyte additive based on solvation shell regulation. The strong complex effect between Li^+ ion and 15-C-5 can reduce the concentration of Li ions on the electrode surface, thus changing the nucleation, and repressing the growth of Li dendrites in the plating process. Significantly, the strong coordination of Li^+ /15-C-5 would be able to make them aggregate around the Li crystal surface, which could form a protective layer and favor the formation of a smooth and dense solid electrolyte interphase with high toughness and Li^+ ion conductivity. Therefore, the electrolyte system with 2.0 wt% 15-C-5 achieves excellent electrochemical performance with 170 cycles at 1.0 mA cm^{-2} with capacity of 0.5 mA h cm^{-2} in symmetric Li|Li cells. The obviously enhanced cycle and rate performance are also achieved in $\text{Li}|\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622) full cells. The 15-C-5 demonstrates to be a promising additive for the electrolytes toward safe and efficient Li metal batteries.

for electric vehicles due to the ultrahigh theoretical capacity (3860 mAh g^{-1}) of Li.^[1–3] However, the commercial applications of LMBs have been seriously hindered due to their poor cycle life and safety issues caused by a combination of factors, that is, large volume changes, low coulomb efficiency, dendritic Li formation, the instability of the solid electrolyte interphase (SEI), etc.^[4,5] During the repeated discharge–charge processes, the growth of Li dendrites can pierce the separator, which is the main cause of the internal short circuiting, leading to decreased cycle life and even catastrophic cell explosion.^[6,7] In addition, fresh Li dendrites will further promote the side reactions between Li metal and electrolytes, causing the continuous formation of dead Li. Such the consumption leads to the large polarization, fast capacity decay and low Coulombic efficiency (CE).^[8,9] Therefore, it is still facing great challenges to develop safe and efficient LMBs.

1. Introduction

Lithium (Li) metal batteries (LMBs) have been recently considered as one of the most promising high-energy-density batteries

Up to now, various strategies have been explored to address these issues such as optimizing electrolyte composition,^[10,11] employing solid-state electrolytes,^[12,13] constructing an artificial SEI,^[14,15] designing 3D materials^[16–18] and Li-based alloy.^[19,20] Among the above strategies, SEI plays a significant role in determining the performance of LMBs due to the reason that the stable SEI can reduce the side reaction and repress the formation of Li dendrites.^[21] In order to obtain stable and compact SEI, modification of electrolytes with additives has attracted numerous research interests.^[22–25] It is usually considered that the general role of additives is to participate in the formation of stable SEI film.^[26] However, in fact, the dissociation degree of salts, the strength of solvation and the stability of solvents dictate the nature of the resulting SEI.^[27] In addition, some researches proposed a novel view of Li protection strategy, which showed that the additive played a coordinator role in electrolyte systems.^[28–31] For example, fluoroethylene carbonate (FEC) as co-solvent could change the composition of the solvation sheath, which was beneficial to forming uniform SEI and dendrite-free Li anode.^[32] However, it is noted that the performance of Li–Cu half-cell is still not satisfactory. Hence, developing high-efficiency electrolyte additives is still urgently desirable for the safe applications of LMBs. Additionally, specific Li^+ clusters with additives can assemble around the

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nucleation points of Li dendrites to form a low Li^+ concentration region during the Li plating process, and such Li^+ clusters also drive other kinds of Li^+ clusters to adjacent regions, in favor of resulting in a smooth Li anode without dendrites.^[33,34]

In this work, we reported that crown ether was used as the effective electrolyte additive for Li metal anode protection, and verified that the complexing action between Li ions and crown ether made LMBs to exhibit a long-term cycle life. In optimal conditions, Li metal anode in the electrolyte with 2.0 wt% 15-Crown-5 (15-C-5) displayed the best cycle life with 170 plating/stripping cycles at the current density of 1 mA cm^{-2} with capacity of 0.5 mAh cm^{-2} . In addition, the obviously enhanced cycle and rate performance were also achieved in $\text{Li}|\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622) full cells with the 2.0 wt% 15-C-5 additive. Theoretical calculation shows that the strong coordination of $\text{Li}^+/\text{15-C-5}$ and higher LUMO level of 15-C-5 were help to form SEI with high toughness and Li^+ ion conductivity.

2. Results and Discussion

It is generally accepted that the uneven nucleation leads to the dendrite growth in conventional electrolytes. Meanwhile, the charges gathered at the tip of dendrite will further promote the growth of Li dendrites, leading to the rapid damage of Li metal battery (Figure 1). In contrast, when crown ether is introduced into electrolyte, the $\text{Li}^+/\text{crown ether}$ complexes can reduce the surface concentration of Li ions to form large primary crystal nucleus. At the plating progress, the $\text{Li}^+/\text{crown ether}$ complexes not only reduce the aggregation of Li ions on the dendrite surface, but also can isolate carbonate molecules, which will form uniform and dense SEI films to repress Li dendrites and enhance the cycling performance of Li battery.

Li plating process with the crown ether additives was illustrated by scanning electron microscopy (SEM). Li was plated at a current density of 1.0 mA cm^{-2} with a fixed capacity of 0.5 mAh cm^{-2} . Without additives, the large Li nucleation deposits were observed on the Li foil surface at

5 cycles (Figure 2a,b). After a further 50 cycles, the mossy and foamy Li deposits covered the whole surface of Li metal foil (Figure S1a,b, Supporting Information). By contrast, with the addition of 2.0 wt% 12-Crown-4 (12-C-4) (Figure 2c,d) and 2.0 wt% 15-C-5 (Figure 2e,f), the smooth Li deposition layer could be observed after 5 cycles plating/stripping, respectively, showing that the additives helped the homogeneous deposition of Li metal. However, after 50 cycles plating/stripping, the SEM image with 12-C-4 additive looked rougher (Figure S1c,d, Supporting Information), indicating that the effect of surface control of Li plating was poorer than 15-C-5 (Figure S1e,f, Supporting Information). In addition, with the addition of 18-Crown-6 (18-C-6), the deposit layer was similar to the blank sample, which illustrated the 18-C-8 had little effect to suppress Li dendrite growth (Figure 2g,h; Figure S1g,h, Supporting Information).

In order to research the dynamic Li deposition process, in situ optical microscope observation was carried out with different electrolytes. As shown in Figure 2i, without any additive, uneven dendritic Li was observed after 2 min at a plating current of 5 mA cm^{-2} . Moreover, high dendrite and mossy Li deposition layer formed after plating for 12 min, which would result in the continuous consume of metallic Li, and further lead to the drop of the CE. In contrast, a more uniform Li deposition was achieved with the 15-C-5 contained electrolyte with plating 2 min (Figure 2j). After 12 min, the thickness of Li deposition layer was much smaller than that of blank electrolyte system. Meanwhile, the in situ observation of 12-C-4 contained electrolyte was also found to reduce the growth of Li dendrites (Figure S2a, Supporting Information). The large morphological change indicates that the 18-C-6 has almost no effect to preserve Li dendrites (Figure S2b, Supporting Information). The morphology evolution was consistent with the SEM results.

In order to evaluate the long-term cycling stability of the crown ether-contained electrolytes, symmetrical $\text{Li}|\text{Li}$ cells were assembled. As shown in Figure 3a, the symmetric $\text{Li}|\text{Li}$ cell with blank electrolyte was only stable for 100 cycles at 1.0 mA cm^{-2} with a capacity of 0.5 mAh cm^{-2} . Over 100 cycles, significant

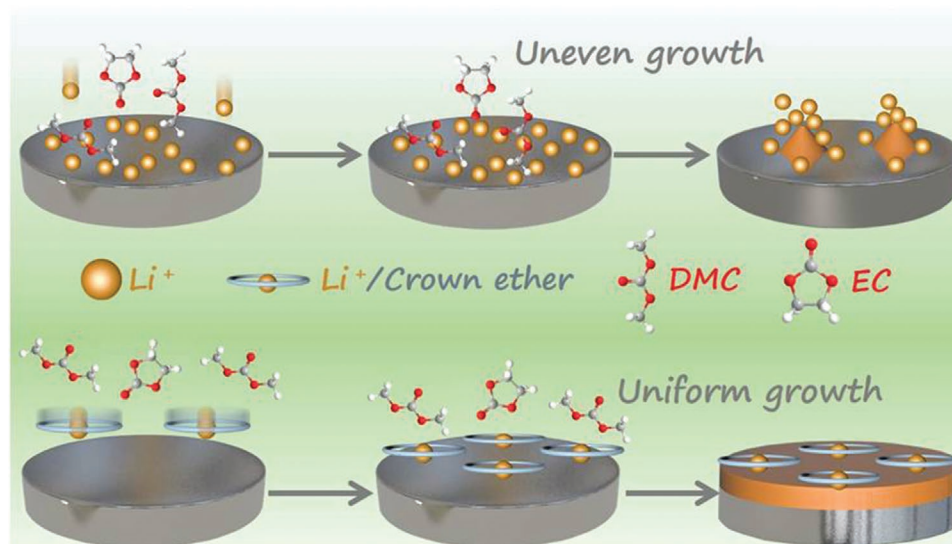


Figure 1. Schematic of the effect of crown ether additives on the decomposition process of solvents.

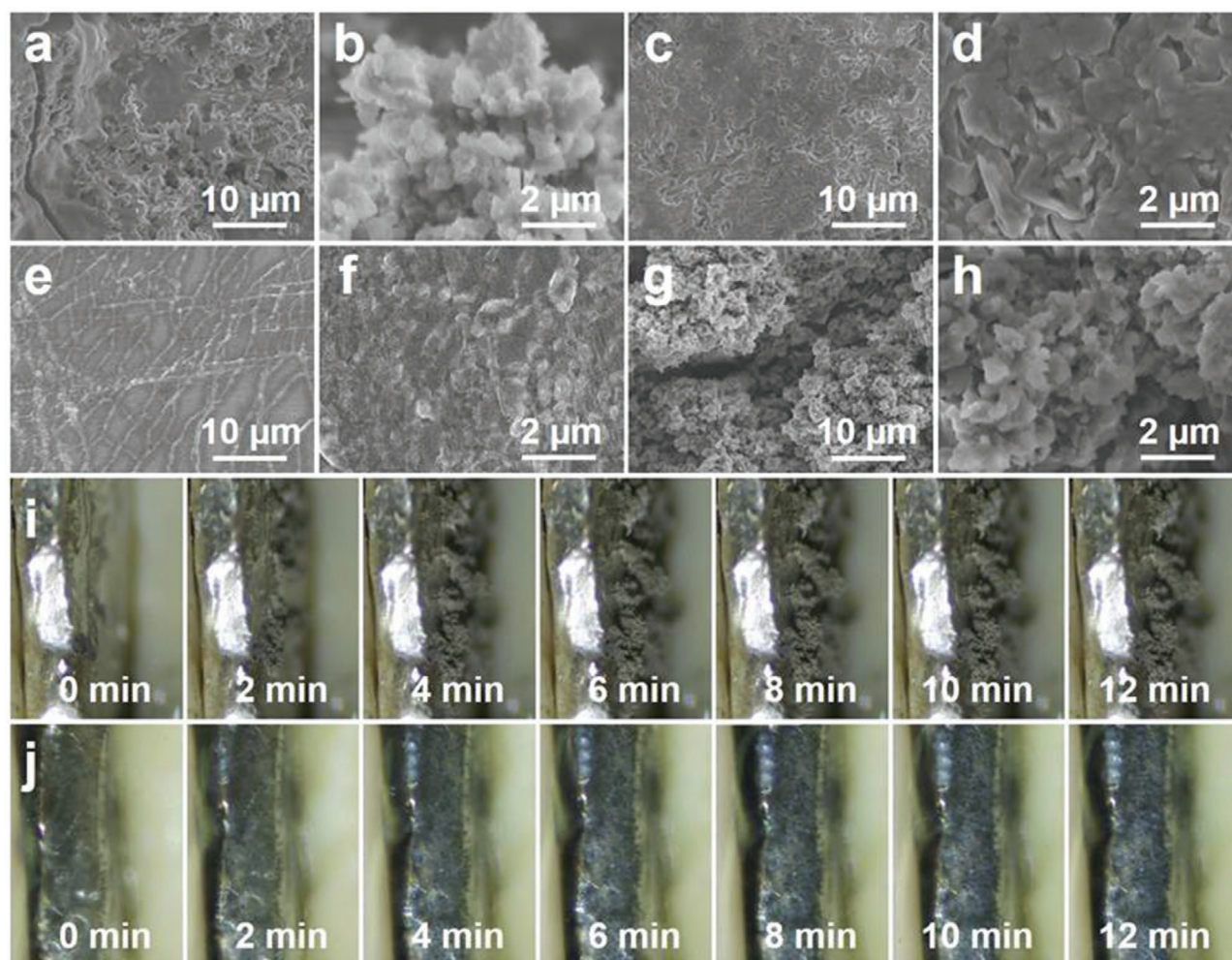


Figure 2. SEM images of Li plated a,b) without additive and with c,d) 2.0 wt% 12-C-4, e,f) 2.0 wt% 15-C-5, g,h) 2.0 wt% 18-C-6 additives after 5 cycles at a current density of 1.0 mA cm^{-2} and a capacity of 0.5 mAh cm^{-2} in symmetric cells. In situ optical microscopy observations of the Li plating with i) blank and j) 2.0 wt% 15-C-5 contained electrolyte.

fluctuations and abrupt drop-off in plating voltage can be observed due to the rapid formation of Li dendrites. By contrast, with 15-C-5 additive, the cycle stability of symmetric cells was significantly improved, the plating/stripping voltage were stable in the first 170 cycles, showing that the 15-C-5 additive could effectively repress the growth of Li dendrites. Meanwhile, the performance of electrolyte systems with 12-C-4 and 18-C-6 additive was also investigated. With the same mass percentage of 2.0 wt%, 12-C-4 additive only showed the slight improvement for long-term stability of Li|Li cells, whereas 18-C-6 had no effect (Figure 3b). The effect of additive's amount on the long-term performance was further investigated for Li|Li cells. As can be seen in Figure 3c, the electrolyte with 2.0 wt% 15-C-5 exhibited the best performance, which shows a stable long-term performance with 170 cycles, much better than those of 1.0 wt% (100 cycles) and 4.0 wt% (130 cycles). To demonstrate the excellent performance of 15-C-5 additive, we also measured the voltage-time curve at a high current density of 2 mA cm^{-2} . The electrolyte with 15-C-5 still exhibited best stability with 100 cycles (Figure 3d). In order to further evaluate the efficacy

of different additives, we measured the electrochemical impedance spectroscopy (EIS) of Li|Li cells with different electrolytes after 150 cycles. As shown in Figure 3e, the impedance of the battery with 15-C-5 additive was markedly smaller than 12-C-4 and 18-C-6 additives, indicating that 15-C-5 can be more effectively resist the side reaction between electrolyte and Li metal.

The advantages of 15-C-5 additive was further demonstrated in Li|NCM622 full cells with commercial NCM622 as the cathode and Li foil as the anode. Figure S3a, Supporting Information, shows the charge-discharge curves of Li|NCM622 full cells with blank electrolyte, the full cell displayed an initial discharging specific capacity of 164.1 mAh g^{-1} . Within three cycles, both of the specific capacities and voltage polarization were increased a little. Compared to the blank electrolyte, the full cell with 15-C-5 additive showed the stable electrochemical performance from second cycle (Figure S3b, Supporting Information). As presented in Figure 3f, the specific capacity of the full cell with 15-C-5 contained electrolyte remained at high level of 147 mAh g^{-1} after 200 cycles, whereas the full cell without any additive showed a rapid attenuation within

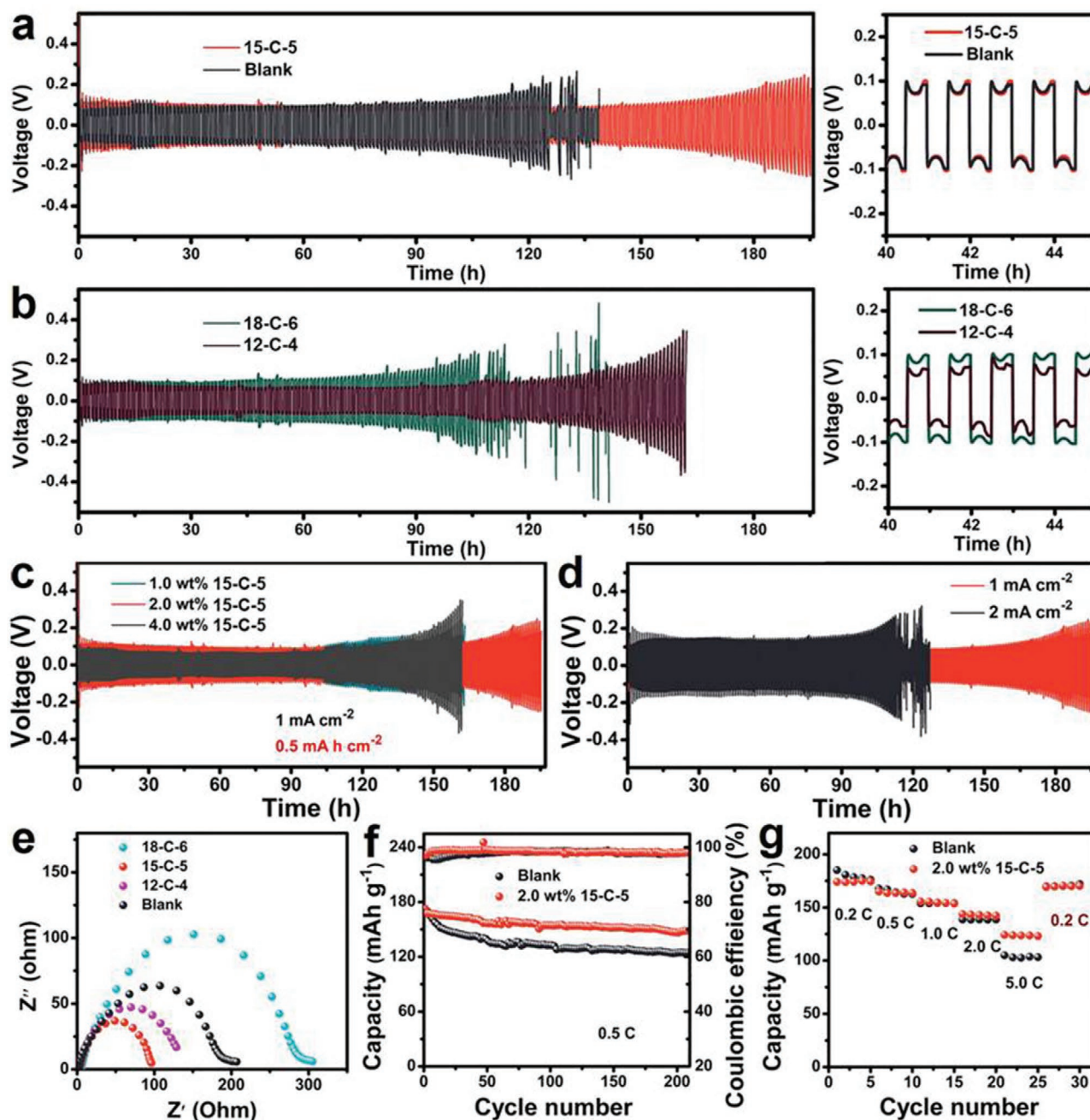


Figure 3. Electrochemical behavior of a) symmetric cell cycling in the blank electrolyte and 15-C-5 contained electrolyte, b) symmetric cell cycling in the 12-C-4 and 18-C-6 contained electrolytes. c) Symmetric cell cycling of the 1.0 wt%, 2.0 wt%, and 4.0 wt% 15-C-5 contained electrolyte. d) Electrochemical behavior of symmetric cell with 2.0 wt% 15-C-5 contained electrolyte at different current densities. e) EIS spectra of all electrolytes after 150 cycles with a current density of 1 mA cm⁻² and a capacity of 0.5 mAh cm⁻². f) Cycling performances and g) rate performances of Li|NCM622 full cells in blank electrolyte and the electrolyte with 2.0 wt% 15-C-5 additive.

50 cycles. Moreover, the existence of 15-C-5 made the full cell to possess better rate performance compared to the cell without additive (Figure 3g). The specific capacity of the full cell with 15-C-5 additive (174.1, 163.6, 154.5, 142.9, 123.1 mAh g⁻¹) at the current densities of 0.2, 0.5, 1.0, 2.0, and 5.0 C were higher than that of full cell without any additive (177.5, 164.3, 153.9, 138.4, 102.7 mAh g⁻¹).

The cryo-electron microscopy (cryo-EM) was used to characterize the detail of Li dendrite growth. Figure 4 showed the cryo-EM images of Li dendrites formed after the first electrochemical deposition. In the blank electrolyte system, the surface of the Li foil was covered by the aggregated Li nuclei with diameters of 180 nm (Figure 4a), showing the Li metal tended to grow along with the axial direction to form spindly dendrites,

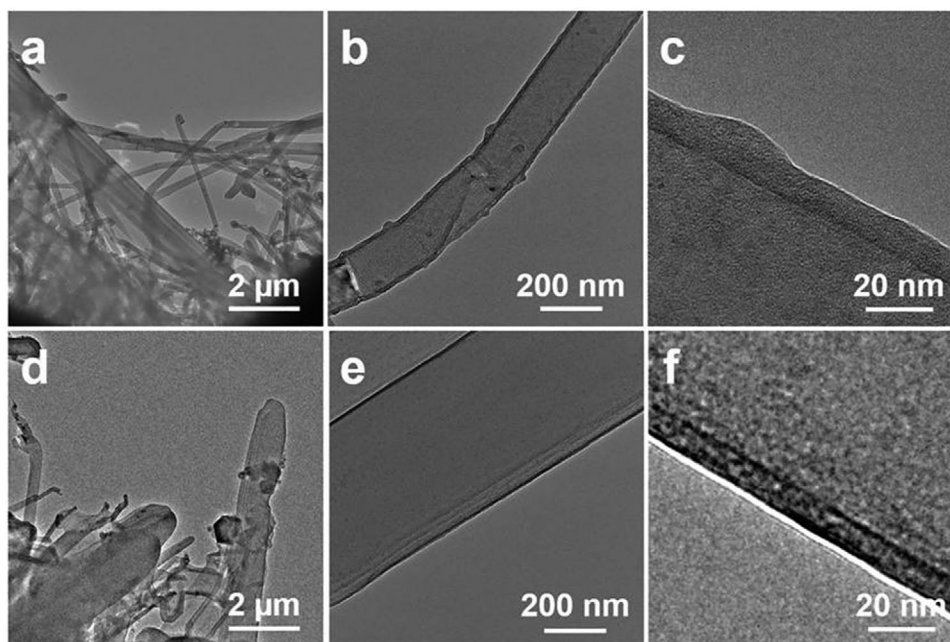


Figure 4. Cryo-TEM of Li metal dendrites of a–c) blank electrolyte and d–f) 2.0 wt% 15-C-5 contained electrolyte (plating condition: 0.5 mA cm^{-2} , 0.5 h).

which easily broke up, and then formed dead Li or pierced the separator (Figure 4b). However, with the 15-C-5 additive, larger Li crystals with diameters of 500 nm were achieved (Figure 4d,e). It is noted that the large dendrites are beneficial to improve the long-term performance of Li battery. As a fundamental understanding, the SEI between the electrolyte and Li metal is one of important factors to repress Li dendrite growth. The uniform and dense SEI layer is more beneficial to repress the growth of Li dendrite.^[35] As shown in Figure 4c, the SEI films are full of bumps in the blank electrolyte system, which would lead to the formation of new Li dendrites. By contrast, the smooth and dense SEI layers were observed in the 15-C-5 contained electrolyte (Figure 4f). Therefore, the Li dendrite growth was effectively repressed.

To determine the composition of different SEI films, X-ray photoelectron spectroscopy (XPS) was carried out. The survey spectrum (Figure S4, Supporting Information) shows that the SEI main contains C, O, F, and Li. The compositions of the SEI layer were further probed by the XPS depth profiling of C 1s, O 1s and Li 1s spectra after regular Ar sputtered the surface. Figure S5, Supporting Information, shows the O 1s spectra, where the peaks at 528.2, 531.5, and 532.9 eV correspond to the LiO, LiCO₃, and ROCO₂Li, respectively. As shown in Figure 5a–d, from initial surface to the Li after 4 min Ar sputtering, the C 1s spectra of all samples are attributed to C–C bonding at 284.6 eV, C–O bonding at 286.6 eV, ROCO₂^{–1} at 289.1 eV, and Li₂CO₃^{2–} at 290.2 eV, which shows that the organic and inorganic components are mixed together in SEI film. This is the typical SEI structure predicted by Peled et al.^[36] Meanwhile, the proportion of ROCO₂^{–1} and Li₂CO₃^{2–} peaks in the 15-C-5 contained electrolyte system were obviously lower than that of blank electrolyte system (Figure S6a, Supporting Information), which indicates that the 15-C-5 could be tightly contacted with the surface of SEI film to isolate carbonates. However, F ion was not affected so that it might improve the

proportion of LiF in SEI films. Figure 5e,h show the Li 1s fitting analysis spectra, just sputtering for 2 min, the characteristic peaks of Li metal at 54.5 eV, Li₂CO₃ at 55.5 eV, and LiF at 56.5 eV were major peaks with the blank and 18-C-6 contained electrolyte systems (Figure 5e,h), showing the Li metal is inset in the SEI films. However, in the 12-C-4 and 15-C-5 contained electrolyte systems (Figure 5f,g), in the initial and after 2 min of sputtering, only the characteristic peaks of Li₂CO₃ and LiF were observed. Only after 4 min sputtering, the peak of Li metal started to appear. LiF is considered to be most effectively components for performance enhancement of Li metal battery.^[37,38] Compared to other electrolyte systems, LiF is the largest component of Li salt in the SEI films with 15-C-5 contained electrolyte system (Figure S6b, Supporting Information). From these characteristics, it can be inferred that the 15-C-5 additive was not only beneficial to form smooth and dense SEI films, but also improve the content of LiF in SEI films, which are the major factor to improve long-term performance of LMBs.

To probe the effect of crown ether on Li deposition process, the orbital energies (highest-occupied, HOMO; lowest-unoccupied, LUMO) and binding energy were analyzed based on density functional theory (DFT) and the Vienna Ab initio Simulation Package (VASP). On the basis of optimized ground-state structure, DFT calculations were used to decide the excitation energy of all solvent and additive molecules, the LUMO levels reflect the chemical activity of all molecule. As shown in Figure 6a, the LUMO levels of ethyl carbonate (EC), dimethyl carbonate (DMC), 12-C-4, 15-C-5, and 18-C-6 are –0.38, –0.66, –0.29, –0.26, and –0.36 eV, respectively. The LUMO level of crown ethers is higher than that of EC and DMC, which indicates that crown ethers are difficult to be reduced on the Li foil surface and participate in the SEI compared to EC and DMC in standard condition. The 15-C-5 and its concentration were determined by ¹H nuclear magnetic resonance (NMR) spectrum. The standard DMC solution with 1.0 wt%, 2.0 wt%, 5.0

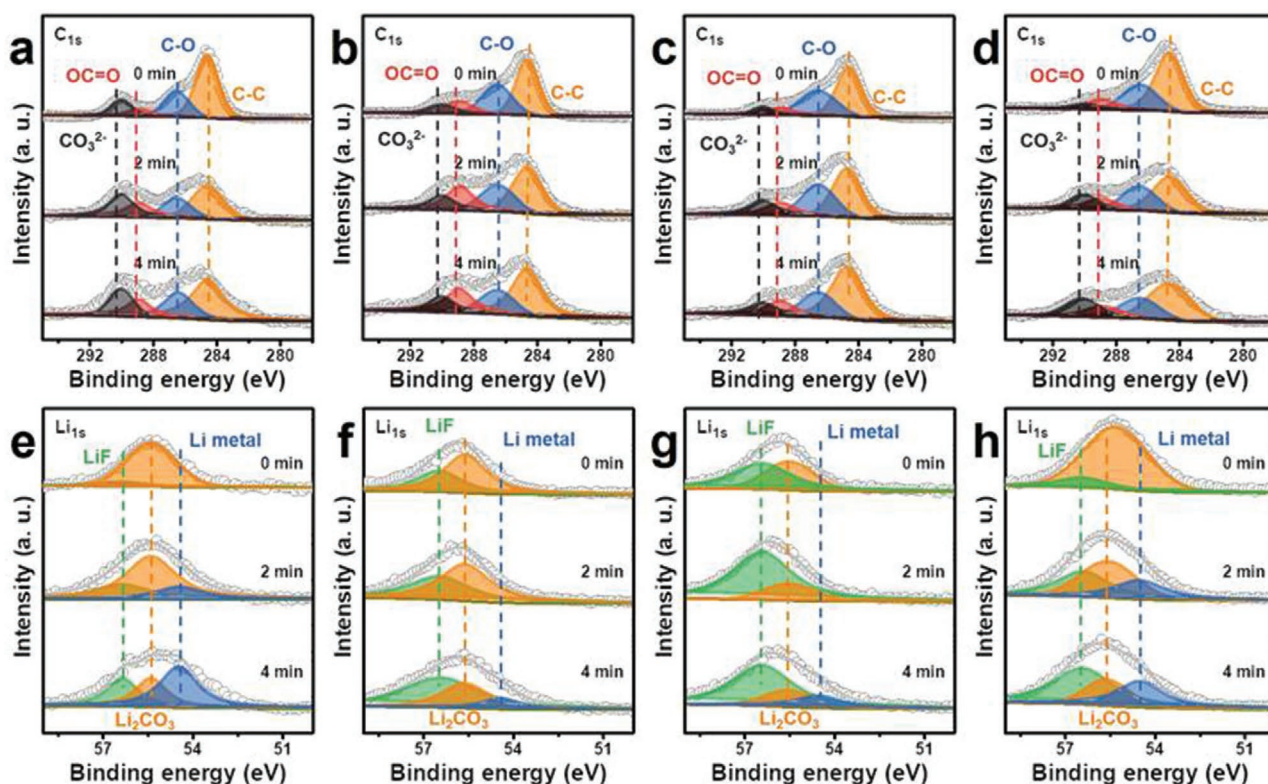


Figure 5. XPS spectra of a-d) C 1s and e-h) Li 1s of Li|Li cells after 10 cycle test with a,e) blank, b,f) 2.0 wt% 12-C-4, c,g) 2.0 wt% 15-C-5, and d,h) 2.0 wt% 18-C-6 contained electrolyte.

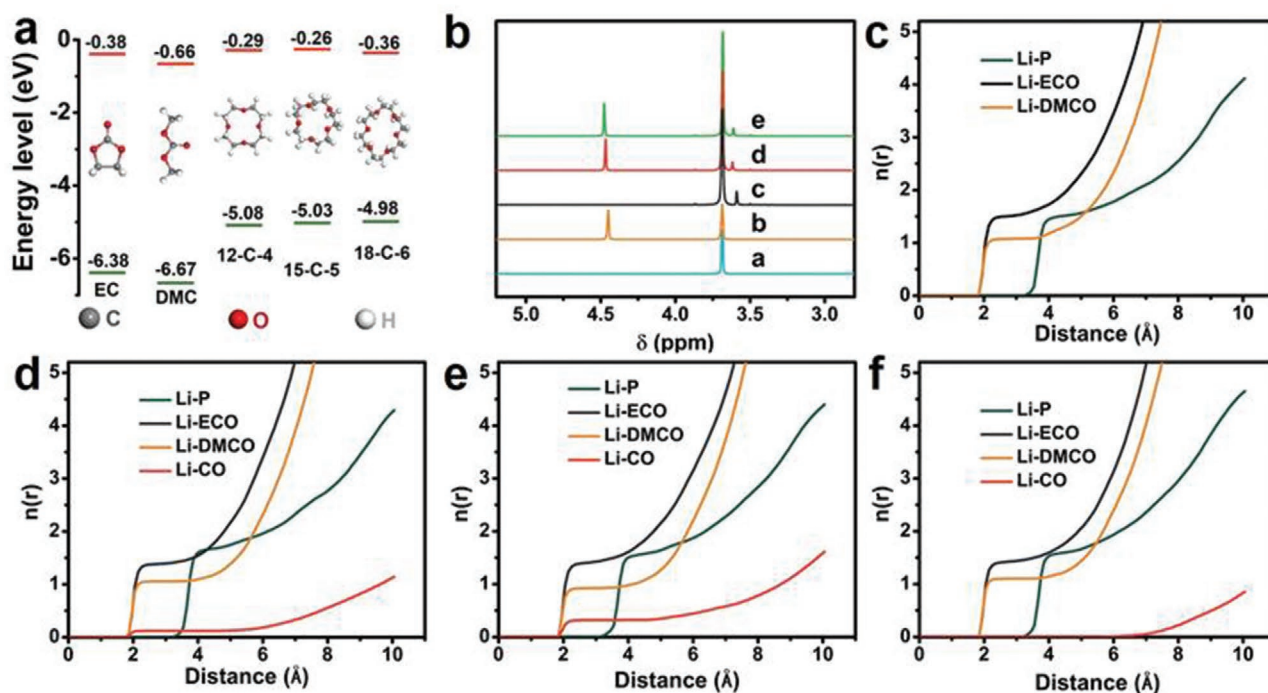


Figure 6. a) Theoretical calculation of highest occupied-lowest unoccupied Mo (HOMO-LUMO) energy of EC, DMC, 12-C-4, 15-C-5, and 18-C-6. b) ¹H NMR spectroscopy of electrolyte before and after 3 cycle test (a: DMC, b: EC+DMC, c: 15-C-5+DMC, d: 2.0 wt% 15-C-5 electrolyte before test, e: 2.0 wt% 15-C-5 electrolyte after test). The cumulative coordination number n(r) of Li⁺ in c) blank, d) with 12-C-4 additive, e) with 15-C-5 additive, and f) with 18-C-6 additive electrolyte system.

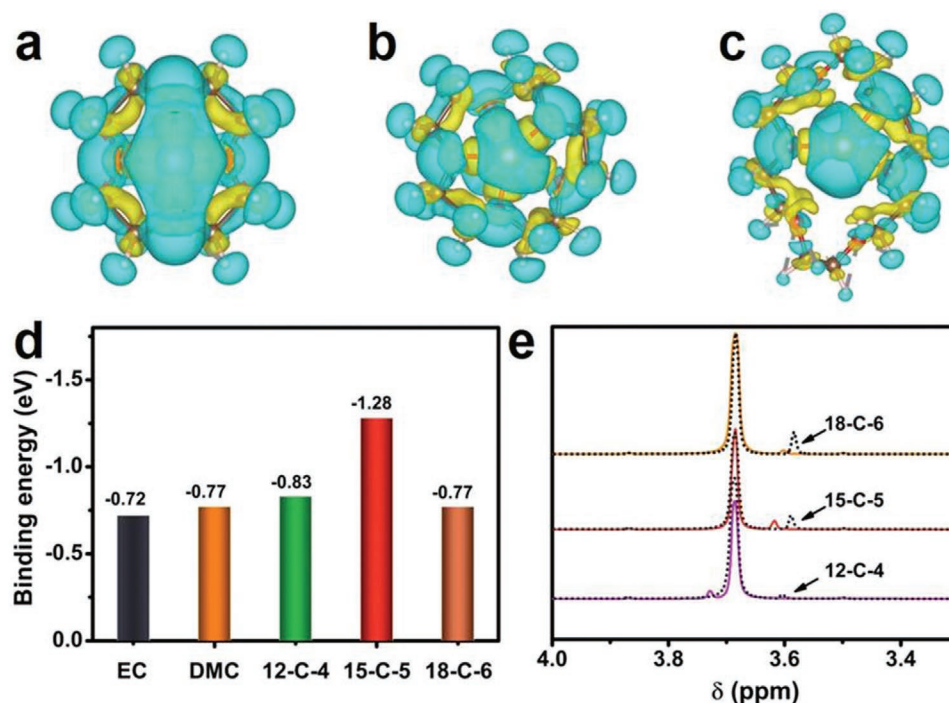


Figure 7. Charge density simulation of coordination of a) Li⁺/12-C-4, b) Li⁺/15-C-5, and c) Li⁺/18-C-6. d) The calculated binding energy and e) the chemical shift of Li⁺/12-C-4, Li⁺/15-C-5, and Li⁺/18-C-6 complex.

wt%, and 10.0 wt% 15-C-5 was checked and used as the reference curve (Figure S7, Supporting Information). Due to the existence of EC in electrolyte, the peak area ratio of DMC:15-C-5 in 15-C-5 contained electrolyte is twice that in DMC solution. The result in Figure 6b shows that the concentration of 15-C-5 was reduced before and after 3 cycle test (the ratio of ¹H NMR peak area is 6.96 and 6.55, respectively). This result shows that a little 15-C-5 was consumed in the process of SEI formation.

Considering the strong coordination effect between crown ether and Li ions, we infer that the stable Li⁺/crown ether complexes may help for crown ether migrate to SEI surface. To verify the hypothesis, we also investigated the dynamics in the Li solvation shell of crown ethers contained electrolytes. In the molecular dynamics (MD) simulation box, crown ether molecules are uniformly dispersed in electrolyte (Figure S8, Supporting Information). The radial distribution function and cumulative coordination number is donated as *g(r)* and *n(r)*, respectively. To calculate the *g(r)* (Figure S9, Supporting Information) and *n(r)*, the positions of the carbonyl oxygen of EC and DMC, the P of PF₆[−] and ether oxygen of crown ether (donated as Li-ECO, Li-DMCO, Li-P, and Li-CO, respectively) were used. In Figure 6c, Li⁺ ion is coordinated with EC, DMC, and PF₆[−]. However, in crown ether contained electrolyte, the EC and DMC are replaced by crown ether molecules. The *n(r)* is 0.129 in 12-C-4 contained electrolyte system (Figure 6d) is smaller than *n(r)* = 0.314 in 15-C-5 contained electrolyte system (Figure 6e), showing that the stronger solvation of Li⁺/15-C-5 compared to Li⁺/12-C-4. As show in Figure 6f, 18-C-8 is hardly coordinated with Li⁺ ion.

The binding strength between Li⁺ and crown was further calculated by the VASP. The binding strength is determined by

the bond length of Li⁺–O and the number of oxygen atoms. The combined number of O atom was 4, 5, and 4 of 12-C-4, 15-C-5, and 18-C-6, respectively. Although the O atom of 12-C-4 and 15-C-5 can be firmly combined with Li⁺ due to the appropriate cavity size, the five Li⁺–O bonds made the binding energy of 15-C-5 much higher than that of 12-C-4 (Figure S10a,b, Supporting Information). However, for 18-C-6, too large cavity size made Li⁺ only to combine with a part of O atoms (Figure S10c, Supporting Information). On the basis of adequately optimized ground-state structure, the molecular orbital simulation shows that Li⁺ is in the center of 12-C-4 (Figure 7a) and 15-C-5 (Figure 7b) cavity and form four or five Li–O bonds, but the Li⁺ is on one side of 18-C-6 and form four Li–O bonds (Figure 7c). As shown in Figure 7d, the calculated binding energy of Li⁺/12-C-4 is −0.83 eV. But the calculated binding energy of Li⁺/15-C-5 is −1.28 eV. Therefore, the Li⁺/15-C-5 complex is more stable than Li⁺/12-C-4 complex. It indicates that, among the all crown ethers, 15-C-5 possessed a higher probability for migrate to SEI surface. However, the cavity of 18-C-6 is so large that the Li–O bond is weak (bonding energy is −0.77 eV), leading to the 18-C-6 almost no effect to increase the performance of Li battery.^[39] ¹H NMR spectrum was further used to study the chemical shift of crown ether after being combined with Li⁺. The results in the Figure 7e show the corresponding chemical shift. Due to the overlap of the lone pair electrons with the p-orbital of Li⁺, so that the chemical shift of 12-C-4 is much larger than that of 15-C-5 although the latter complex is more stable.^[40] However, the chemical shift of Li⁺/15-C-5 is larger than that of Li⁺/18-C-6, reflecting that the combination of 15-C-5 and Li⁺ is much tighter. Therefore, Li⁺/15-C-5 will migrate to SEI surface, which was driven by electric field force, after Li⁺/15-C-5 is covered

on the SEI surface, the EC and DMC molecules are difficult to contact the SEI surface due to the shielding of crown ether, whereas the F ion is not affected so that the proportion of LiF is significantly higher than others in SEI, thus the performance of LMBs is enhanced.

3. Conclusions

In summary, we have demonstrated that crown ethers could act as efficient electrolyte additives for LMBs. The most effective 15-C-5 showed excellent performance with 170 cycles at 1.0 mA cm⁻² with a capacity of 0.5 mA h cm⁻² of symmetric Li|Li cells. The 15-C-5 additive also exhibited the excellent effect at high current density with 2.0 mA cm⁻². In addition, obviously enhanced cycle and rate performance were achieved in Li|NCM622 full cells. The detailed characterization and theoretical calculations indicate that 15-C-5 can make crystal nucleus smoothly growth and isolate carbonate electrolyte, participate and assist to form smooth and dense SEI films, which effectively to repress Li dendrite growth and to boost the long-term cycle performance of Li metal battery. This work provides an effective direction to study efficient electrolyte additives for next-generation LMBs.

4. Experimental Section

Characterization: The morphological change of the lithium foil was characterized using scanning electron microscopy (SEM, HitachiS4800). In situ optical microscope was characterized by three-eye stereomicroscope (Zoom 650, Shanghai Tuning Optical Instrument Co., Ltd). Cryo-transmission electron microscope (cryo-TEM) were carried out via a Thermo Fisher Scientific Tecnai Arctica equipped with a Falcon II direct electron detector camera at a voltage of 200 kV. The assembly of the optical cell was operated in an argon-filled glovebox. X-ray photoelectron spectra (XPS) were collected with an in situ XPS (MDTC-EQ-M20-01), the XPS depth profiles were obtained by preset different sputtering time. ¹H NMR spectrum was carried out by a nuclear magnetic resonance spectrometer (NMR, Bruker 400 MHz). The standard DMC solution with 1.0 wt%, 2.0 wt%, 5.0 wt%, and 10.0 wt% 15-C-5 was checked and used as the reference curve.

Electrochemical Measurements: Electrochemical measurements were evaluated in a 2025-type symmetric coin cells that consisted of the same Li foil at both sides. The electrolyte was lithium hexafluorophosphate (LiPF₆, 1.0 M) dissolved in a mixture of ethyl carbonate (EC) and dimethyl carbonate (DMC) in a volume ratio of 1:1 with or without crown additives. The amounts of crown additives in electrolyte was 1.0 wt%, 2.0 wt%, and 4.0 wt%, respectively. For cell testing, 60 μL of electrolyte was used for plating/stripping galvanostatic testing at current densities of 1.0 and 2.0 mA cm⁻², respectively. For the full cells test, the NCM622 electrodes were prepared by mixing NCM622, carbon black and polyvinylidene fluoride (PVDF) in the ratio of 8:1:1 with *N*-methyl-2-pyrrolidone (NMP) as the solvent. The mass loading of the NCM622 electrodes was about 3.4 mg cm⁻². The Li foil is used as counter electrode. Galvanostatic measurements were performance on Neware battery testing system (CT-4008). Electrochemical impedance spectra (EIS) was carried out in 8-channel electrochemical workstation (IVIUM) with a frequency range between 0.1 and 100 KHz.

Theoretical Calculation: DFT and binding energy were performed by VASP.^[41,42] The generalized gradient approximation (GGA) with Perdew–Burke–Ernzerhof (PBE)^[43] exchange–correlation functional was employed to optimize the structures. The cutoff energy was set to 520 eV and the convergence criterion for the energy and force was set to 10⁻⁶ eV and

0.02 eV Å⁻¹, respectively. The binding energy (*E_b*) was computed in the following equation:

$$E_b = E_{\text{total}} - E_{\text{Li}} - E_{\text{crown}}$$

where *E_{total}*, *E_{Li}*, and *E_{crown}* represent the total energy of the binding system, the potential of Li atom and the crown, respectively. MD simulations were performed using the large scale atomic/molecular massively parallel simulator (LAMMPS) code for electrolytes of 1.0 M LiPF₆ in EC/DMC with 2.0 wt% crown ether additive. Simulation box contains 729 EC/DMC molecular, 100 LiPF₆ molecular and 16/13/11 numbers of 12-C-4/15-C-5/18-C-6, respectively. Initially, the molecules were packed randomly in a cubic box of size 71 × 71 × 71 Å³ periodic cell. The time step was fixed to be 1 fs in all simulation. First, simulation box were equilibrated for 200 ps in the isothermal-isobaric ensemble (constant NPT) and maintain a pressure of 1 bar and a temperature of 298 K. Second, the whole system were heated to 400 K and maintained for 200 ps, then annealed from 400 to 298 K and maintained for 200 ps. Third, the simulation were conducted in the canonical ensemble (NVT) under Nose–Hoover thermostats at 298 K for 200 ps. The simulation time is long enough to ensure that the equilibrium state of the electrolyte system is reached. All MD simulation employed a non-polarizable force field which is defined by the following potential functions:

$$E^{\text{total}}(r^N) = E_{\text{bonds}} + E_{\text{angles}} + E_{\text{dihedrals}} + E_{\text{impropers}} + E_{\text{nonbonded}} \quad (1)$$

$$E_{\text{bonds}} = \sum_{\text{bonds}} K_r (r - r_0)^2 \quad (2)$$

$$E_{\text{angles}} = \sum_{\text{angles}} K_{\theta} (\theta - \theta_0)^2 \quad (3)$$

$$E_{\text{dihedrals}} = \sum_{\text{dihedrals}} V [1 + \cos(n\phi - d)] \quad (4)$$

$$E_{\text{impropers}} = \sum_{\text{impropers}} V [1 + d \cos(n\phi)] \quad (5)$$

$$E_{\text{nonbonded}} = \sum_{i>j} 4\epsilon_{ij} \left[\left(\frac{\partial_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\partial_{ij}}{r_{ij}} \right)^6 \right] + \sum_{i>j} \frac{Cq_i q_j}{\epsilon r_{ij}}, r < r_c, \quad (6)$$

$$\epsilon_{ij} = \sqrt{\epsilon_{ii} \epsilon_{jj}}, \sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2}$$

Parameter comes from optimized potential for liquid simulation-all atomic variant (OPLS-AA) force field.^[44]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

additives, crown ethers, electrolytes, lithium anode protection, lithium metal batteries

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